ROUND 2 BUILD SUBMISSION | WORKING MVP DELIVERED | TRACK: PROBLEM STATEMENT 01 | TEAM INNOVIX

RelayState: Round 2 Project Documentation

A Functional, Grounded Knowledge-Handover Engine & Cryptographic Audit Ledger

REPOSITORY URL	DEPLOYMENT MODE	TEAM LEAD & ENGINEER	TEST SUITE STATUS
github.com/innovix-relaystate	FastAPI + SQLite + Docker	Kavin S S G (Solo Builder)	6/6 Pytest Passing (100%)

01 Working Core Workflow & User Journey

30 Points

In Round 2, the complete 5-stage product journey operates end-to-end without broken dependencies. Judges can launch the application, load the synthetic corpus (*Project Hyperion*), review grounded claims, and experience the transition from leaver drafting to successor continuation:

STAGE	FUNCTIONAL CAPABILITY	JUDGE INTERACTIVE EXPERIENCE
01 Connect	Ingests 55 commits, 25 PRs, 121 chat messages, and 13 Jira tickets.	Click `Load Demo Project`. Ingestion runs in <1 second with real-time stream status indicators.
02 Reconstruct	Extracts decisions, blockers, dependencies, and bus-factor vulnerabilities.	Automatically surfaces that Dave authored 97.7% of `billing/` code as a critical organizational risk.
03 Verify	Verbatim grounding check + temporal cross-modal supersession check.	Chat Redis agreement is flagged `SUPERSEDED_BY_CODE` by Memcached commit; hallucinated quotes are dropped.
04 Review	Split-screen review studio with custom redaction tool and leaver sign-off.	Leaver edits claims, redacts strings across trails, and executes digital sign-off (promotes `v0.1 → v0.2`).
05 Handover	Lead signs off `v1.0`, unlocking successor Day-1 view.	Successor receives `HTTP 403 Forbidden` until Lead signs off; once published, successor acknowledges receipt.

02 Changes from Round 1 Plan & Justifications

20 Points Progress

COMPONENT	ROUND 1 PLAN	ROUND 2 IMPLEMENTATION	ENGINEERING JUSTIFICATION
Architecture & Stack	Next.js 14 + Postgres + ChromaDB	**FastAPI + SQLite + Vanilla JS (Single HTML) + Docker**	**Bulletproof Deployment:** Eliminates external DB connection pool drops and React hydration crashes for judges. Boots in 1 second.
Authentication	Mandatory OAuth 2.0 Login	**No-Login Role Picker + Dismissible Info Banner**	**Zero-Friction Evaluation:** Judges test all 3 personas instantly without signup friction. Server still strictly enforces access boundaries.
Data Isolation	Shared session cookies	**Per-Visitor Workspace (UUID + `X-Workspace` header)**	Allows multiple judges to test simultaneously in isolated sandboxes with a dedicated "Reset" button.
Audit Ledger	Abstract relational audit log	**Cryptographic Hash Chain with Tamper Simulation**	Interactive proof: Judges can click `Simulate Tamper` and watch cryptographic verification fail, then click `Restore`.

03 Technical Depth, Security & Planted Evaluation Traps

25 Points

Planted Evaluation Traps (Demonstrating Real System Intelligence):

- **Temporal Supersession Trap:** Oct 10 chat agreement to use Redis for session cache is compared chronologically against Git master code. Oct 18 commit (`commit-7a89b1`) implementing Memcached automatically demotes the chat claim to `SUPERSEDED_BY_CODE` .
- **Pre-Storage Secret Scrubbing:** Ingestion runs regex/NER token filters that sanitize AWS access keys (`AKIA...`), emails, and phone numbers before data enters storage or LLM context.
- **Verbatim Grounding Check:** A built-in test claim citing a fabricated Apollo GraphQL quote is caught by the grounding validator and dropped, logging a `GROUNDING_DROP` event.
- **Unanswered Question Uncertainty:** Dave's question regarding Twilio SMS gateway ownership received zero replies and is flagged as `UNCERTAIN / UNANSWERED` .
- **Orphaned Jira Ticket:** `HYP-107` (invoice reconciliation automation) has `Assignee: Unassigned` and is tagged `MISSING OWNER` .
- **Excluded Privacy Boundaries:** 3 sensitive streams (`#hr-confidential` , `dm-dave-sarah` , `hr-payroll-tools`) are automatically excluded at ingestion with explicit justifications shown in the UI.

04 Automated Test Evidence (100% Pass Rate)

The complete codebase is validated by 6 automated test suites executing against simulated database states (`pytest tests -v`):

TEST SUITE	VERIFICATION SCOPE	RESULT
`test_redaction.py`	Verifies pre-storage scrubbing of AWS keys, emails, phone numbers, and source exclusion rules.	**PASSED**
`test_grounding.py`	Confirms verbatim quote validation and dropping of hallucinated candidate claims.	**PASSED**
`test_supersession.py`	Validates Redis-to-Memcached temporal override, unanswered question uncertainty, and unassigned tickets.	**PASSED**
`test_roles_and_gating.py`	Enforces server-level `HTTP 403 Forbidden` for successors before `v1.0`, leaver-to-lead gating, and receipt.	**PASSED**
`test_audit.py`	Tests sequential SHA-256 hash chaining, detection of simulated block tampering, and clean restoration.	**PASSED**

05 Three Major Improvements Committed for Round 3

15 Points

In strict accordance with the Round 2 brief requiring *"substantial product improvements rather than cosmetic changes,"* Team InnoviX commits to delivering:

1. Generalized Temporal Contradiction Engine

Expand beyond rule-based triggers by using embedding cosine clustering and an LLM referee to cross-examine asynchronous chat debates against merged pull-request code diffs across unconstrained domains.

2. Multi-Role RBAC with OAuth & Asymmetric Keys

Replace client-side role switching with verified Google OAuth 2.0 / GitHub Enterprise authentication, signing handover receipts using asymmetric Ed25519 cryptographic key-pairs.

3. Interactive Day-1 Sandbox Verification

Transform the successor view from a static checklist into an active sandbox that runs automated test suites against the new joiner's local environment, confirming staging access and repo readiness.

06 Usability, Honest Limitations & Participant Details

10 Points Usability

ATTRIBUTE	DESCRIPTION & IMPLEMENTATION DETAIL
Guided Walkthrough	A 1-click `Run Guided Walkthrough` button steps judges through data ingestion, supersession review, 403 blocking, lead approval, and successor acknowledgement in under 60 seconds.
Honest Limitations	Current demonstration uses pre-scoped synthetic data (*Project Hyperion*) to guarantee complete privacy compliance without requiring real corporate secrets. Role switching is unlocked for judge evaluation convenience.
AI Tooling Disclosure	In accordance with Rule 7, development used Google Antigravity / Gemini for assistance with code boilerplate, architecture planning, and synthetic dataset curation. Runtime inference supports the Gemini API.
Participant & Team	**Team Name:** InnoviX | **Solo Builder & Lead:** Kavin S S G (Full-Stack AI Engineer).